

ARMELLE BALLIAN

Born on 18.12.1995 in Saint-Étienne, France

Currently at: Climatology and the Biosphere, Geo- and Environmental Research Center
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<https://scholar.google.com/citations?user=ald4LewAAAAJ&hl=en>

FIELDS OF SPECIALIZATION

Continental paleoclimate and paleoenvironmental dynamics, surface processes, geomorphology, paleoaltimetry, terrestrial carbonates, paleosols, fault gouges, karst systems, cave weather monitoring, stable isotope geochemistry ($\delta^{13}\text{C}$, $\delta^{17}\text{O}$, $\delta^{18}\text{O}$, $\delta^2\text{H}$), carbonate clumped (Δ_{47}) isotope thermometry.

PROFESSIONAL RECORD

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| 2025–Present | Research Associate
Department of Climatology and the Biosphere
University of Tübingen, Germany <ul style="list-style-type: none">• Cavity ring-down spectroscopy using a Picarro 2140-i water isotope analyzer• Data-model comparison• Cave weather monitoring |
| 2021–2025 | Doctoral Researcher
Senckenberg Biodiversity and Climate Research Centre (SBIK-F)
Frankfurt am Main, Germany <ul style="list-style-type: none">• Stable and clumped isotope measurements using a Kiel IV Carbonate device paired to a MAT253plus IRMS• Stable isotope measurements using a TC/EA paired to a MAT253plus IRMS• Data-model comparison |
| 2020–2021 | Laboratory Technician
Department of Hydrogeochemistry, Laboratoire de Géologie de Lyon
Université Jean Monnet, Saint-Étienne, France <ul style="list-style-type: none">• Ion chromatography using a Metrohm Eco IC ion chromatograph |
| 2018–2020 | Student Assistant
Department of Sedimentary Geology and Quaternary Research
Albert-Ludwigs-Universität, Freiburg im Breisgau, Germany |

EDUCATION

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| 2021–2026 | Ph.D.
Goethe University, Frankfurt am Main, Germany |
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Thesis: *Miocene paleoclimate and paleoaltimetry reconstructions of the European Alps*

Appreciation: magna cum laude

Advisors: Prof. Dr. Andreas Mulch & Ass.-Prof. Dr. Maud Meijers

2017–2020	Master of Sciences (M.Sc.) in Geology Albert-Ludwigs-Universität, Institute of Earth and Environmental Sciences, Freiburg im Breisgau, Germany
2016–2017	Bachelor of Sciences (B.Sc.) in Earth Sciences ERASMUS Albert-Ludwigs-Universität, Institute of Earth and Environmental Sciences, Freiburg im Breisgau, Germany
2014–2016	Bachelor of Sciences (B.Sc.) in Earth Sciences Université Jean Monnet, Saint-Étienne, France

TEACHING EXPERIENCE

Summer Sem. 2026	Lecturer and co-coordinator Environmental Isotope Chemistry (M208), University of Tübingen Climate Dynamics and Climate Modeling (M322), University of Tübingen Lab practicals Environmental Isotope Chemistry (M208), University of Tübingen Excursions (one day field trip) "Excursion to the Swabian Alb - Karst Processes at the Surface" "Excursion to the Swabian Alb - Karst Processes below the Surface"
Winter Sem. 2025/2026	Lecturer and co-coordinator Global Change (M229), University of Tübingen
Summer Sem. 2025	Guest seminar lecture and lab practical (in English and German) Environmental Isotope Chemistry (M208), University of Tübingen Excursion (one day field trip) "Excursion to the Swabian Alb - Karst Processes at the Surface"

STUDENT SUPERVISION

Bachelor thesis

Summer Semester 2026	Jule Christmann (Water stable isotopes and PFAS in the Swabian Alb karst system)
Summer Semester 2025	Judith Baum (Stable isotope compositions in precipitation on the Swabian Alb)

Master thesis

Summer Semester 2026	Desirée Lo Triglia (Tracing isotopes from rain to calcite)
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GRANTS

September 2025 3000€, **Geogenomic Archaeology Campus Tübingen**
Funded project: "EARTH MEMORY - Caves as Memory Archives of Climate and Life"

HONORS AND AWARDS

September 2024 **Student Poster Presentation Award 3rd place**
GeoSaxonia 2024, Dresden, Germany

December 2023 **Outstanding Student Presentation Award (OSPA)**
AGU Fall Meeting 2023, San Francisco, USA

May 2023 **European Consortium for Ocean Research Drilling (ECORD) scholarship**
for attending the Urbino Summer School in Paleoclimate, Urbino, Italy

INVITED SPEAKER

June 2026 Section Geomorphology at GFZ Helmholtz Centre for Geosciences, Potsdam, Germany

October 2023 Section Climate at Centre de Recherche et d'Enseignement en Géosciences de l'Environnement (CEREGE), Aix-en-Provence, France

May 2023 Department of Earth and Environmental Sciences at University of Freiburg im Breisgau, Germany

SERVICE AND PROFESSIONAL MEMBERSHIPS

2026–Present Past Global Changes (PAGES) - Member of the "Hydroclimate2k" working group (<https://pastglobalchanges.org/user/10729>)

2025–Present Geogenomic Archaeology Campus Tübingen (GACT) - Steering committee member (<https://geogenarch-tuebingen.de/leibniz-sciencecampus/>)

2023–2025 Leibniz PhD Network - PhD Representative and Coordinator of the Working Group Communication (<https://leibniz-phd.net/>)

2022–2024 Young Scientists - Local Speaker (www.senckenberg.de/en/career/young-scientists/)

SCIENCE COMMUNICATION

- September 2026 Presentation (in German) of the EARTH MEMORY project during the *Familienfest* at the Schertelshöhle cave
- September 2026 Introduction to the Swabian Alb and its caves as part of an excursion to the Bärenhöhle in the context of the *International Symposium on Radioglaciology*
- April 2026 Presentation (in German) of the EARTH MEMORY project during the certification process of Hohle Fels cave guides

REVIEWER

Global and Planetary Change, ISSN: 0921-8181

Water, EISSN 2073-4441

CONFERENCE CHAIR

- September 2024 Convener at GeoSaxonia 2024, Dresden, Germany
Session: 'Interactions between mountain building, climate and biodiversity'

PARTICIPATION IN WORKSHOPS AND SUMMER SCHOOLS

- September 2026 **"Lehren und Lernen I" – Hochschuldidaktische Grundlagen**
Hochschuldidaktikzentrum (HDZ) Baden-Württemberg, Tübingen
- July 2026 **Career Development through Research Funding**
Graduate Academy University of Tübingen, Tübingen
- May 2026 **IMPACT Workshop – Past abrupt changes, thresholds, and tipping points: Definitions, Identifications, and Context**
PAGES Impact Working Group, online
- November 2024 **Paleontological Field School, Armenia**
organized by the JURASSICA Museum (Porrentruy, Switzerland) and the Institute of the Geological Sciences of the National Academy of Sciences of the Republic of Armenia (Yerevan, Armenia)
- July 2023 **DMG Data Science Short Course**
organized by the University of Frankfurt, Frankfurt am Main, Germany
- July 2023 **Urbino Summer School in Paleoclimate**
organized by ECORD, Urbino, Italy
- June 2021 **'Principles, applications and new frontiers in isotope geochemistry (PAIG)' Summer School**
organized by the Lake Como School, online

PEER-REVIEWED PUBLICATIONS

- [1] **Ballian, A.**, Meijers, M. J. M., Boateng, D., Cojan, I., Mutz, S. G., Ehlers, T. A., and A., Mulch., 2026. Oxygen isotopic composition of Miocene meteoric water ($\delta^{18}\text{O}_w$) in Alpine foreland basins. *Geochemistry, Geophysics, Geosystems*, 27(8), e2025GC012854. <https://doi.org/10.1029/2025GC012854>
- [2] Liu, X., Liu, Y., **Ballian, A.**, Bernecker, M., Fiebig, J., Rao, Y., Ding, L., and Mulch, A., 2026. Low Elevation and Humid Climate in the Eocene Gerze Basin, Central Tibet: Insights from Multi-Proxy Isotopic Analyses. *Global and Planetary Change*, 105547, ISSN 0921-8181. <https://doi.org/10.1016/j.gloplacha.2026.105547>.
- [3] Boateng, D., Mutz, S. G., Methner, K., **Ballian, A.**, Meijers, M. J. M., Mulch, A., and Ehlers, T. A., 2025. Refining paleoelevation estimates of the European Alps by simulating Middle Miocene climate and $\delta^{18}\text{O}$ responses to diachronous surface uplift scenarios. *Global and Planetary Change*, 251, 104808. <https://doi.org/10.1016/j.gloplacha.2025.104808>.
- [4] **Ballian, A.**, Meijers, M. J. M., Cojan, I., Huyghe, D., Bernecker, M., Methner, K., Tagliavento, M., Fiebig, J., and A., Mulch, 2025. Middle Miocene climate evolution in the Northern Mediterranean region (Digne-Valensole basin, SE France). *Climate of the Past*, 21(4), 841-856. <https://doi.org/10.5194/cp-21-841-2025>.
- [5] **Ballian, A.**, Chawchai, S., Miocic, J. M., Charoenchatree, W., Bissen, R., and Preusser, F., 2024. Late Holocene coastal dynamics south of the Chanthaburi estuary, eastern Gulf of Thailand. *Quaternary Research* 117, 19–29. <https://doi.org/10.1017/qua.2023.34>.
- [6] Bernecker, M., Hofmann, S., Davies, A., Tagliavento, M., Meijer, N., **Ballian, A.**, and Fiebig, J., 2023. A robust methodology for triple (Δ_{47} , Δ_{48} , Δ_{49}) clumped isotope analysis of carbonates. *Chemical Geology*, 642(July), 121803. <https://doi.org/10.1016/j.chemgeo.2023.121803>.
- [7] Boateng, D., Mutz, S. G., **Ballian, A.**, Meijers, M. J. M., Methner, K., Botsyun, S., Mulch, A., and Ehlers, T. A., 2023. The effects of diachronous surface uplift of the European Alps on regional climate and the oxygen isotopic composition of precipitation, *Earth Syst. Dynam.*, 14, 1183–1210. <https://doi.org/10.5194/esd-14-1183-2023>.

CONFERENCE CONTRIBUTIONS PRESENTATIONS

- [1] **Ballian, A.**, Racky, M., Maisch, M., Novello, V., Lo Triglia, D., and Rehfeld, K. A regional scale data–model comparison of modern oxygen stable isotopes in precipitation (Swabian Alb, southwest Germany), EGU General Assembly 2026, Vienna, Austria, 3–8 May 2026, EGU26-7765, 2026. <https://doi.org/10.5194/egusphere-egu26-7765>, 2026.
- [2] **Ballian, A.**, Meijers, M. M. J., Cojan, I., Huygue, D., Bernecker, M., Methner, K., Tagliavento, T., Fiebig, J., and Mulch, A.: Miocene (23-13 Ma) Continental Paleotemperatures in the Northern Mediterranean Region within a Global Climate Framework (Digne-Valensole Basin, SE France), poster, 9th International Clumped Isotope Workshop, Stony Brook University, Stony Brook, USA, 25–28 Aug 2024, 2024.
- [3] **Ballian, A.**, Meijers, M. M. J., Cojan, I., Huygue, D., Methner, K., Boateng, D., Mutz, S. G., Kurz, W., Krsnik, E., Zwingmann, H., Rolland, Y., Ehlers, T., Fiebig, J., and Mulch, A.: Stable isotope paleoaltimetry reveals Early to Middle Miocene along-strike elevation differences of the European Alps, EGU General Assembly 2024, oral presentation, Vienna, Austria, 14–19 Apr 2024, EGU24-18901, <https://doi.org/10.5194/egusphere-egu24-18901>, 2024.

- [4] **Ballian, A.**, Meijers, M. M. J., Methner, K., Cojan, I., Huygue, D., Fiebig, J., and Mulch, A.: Miocene (23-13 Ma) Northern Mediterranean continental paleotemperatures within a global climate framework (Digne-Valensole Basin, SE France), oral presentation, AGU Fall Meeting 2023 held in San Francisco, USA, 11–15 Dec 2023, Session: Paleoceanography and Paleoclimatology / Advances in the Miocene Climate Dynamics II Oral, id. PP54A-04., <https://ui.adsabs.harvard.edu/abs/2023AGUFMPP54A..04B/abstract>, 2023.
- [5] **Ballian, A.**, Meijers, M. M. J., Cojan, I., Huygue, D., Methner, K., Boateng, D., Mutz, S. G., Walter, K., Krsnik, E., Zwingmann, H., Rolland, Y., Ehlers, T., Fiebig, J., and Mulch, A.: Early to Middle Miocene paleoaltimetry results from the Central and Eastern European Alps, poster, AGU Fall Meeting 2023 held in San Francisco, USA, 11–15 Dec 2023, id. PP54A-04., <https://ui.adsabs.harvard.edu/abs/2023AGUFM.T51E0159B/abstract>, 2023.
- [6] **Ballian, A.**, Boateng, D., Mutz, S. G., Meijers, M. M. J., Methner, K., Mulch, A.: The Alps Paleoelevation and Paleoclimate Experiment: Reconstructing Eastward Propagation of Surface Uplift in the ALps (REAL), poster, 7th Annual AlpArray and 4D-MB Scientific Meeting, Bad Hofgastein, Austria, 4–6 Oct 2023, <http://dx.doi.org/10.17169/refubium-40986>, 2023.
- [7] **Ballian, A.**, Meijers, M. M. J., Methner, K., Cojan, I., Huygue, D., Fiebig, J., and Mulch, A.: Miocene (23–13 Ma) continental paleotemperature record from the northern Mediterranean region (Digne-Valensole Basin, SE France) within a global climatic framework, poster, 7th Annual AlpArray and 4D-MB Scientific Meeting, Bad Hofgastein, Austria, 4–6 Oct 2023, <http://dx.doi.org/10.17169/refubium-40985>, 2023.
- [8] **Ballian, A.**, Meijers, M. M. J., Methner, K., Cojan, I., Huygue, D., Fiebig, J., and Mulch, A.: A Miocene (23–13 Ma) continental paleotemperature record from the northern Mediterranean region (Digne-Valensole Basin, SE France) in a global climatic context, poster, MioMEET 2023, Utrecht, Netherlands, 5–7 June 2023, 2023.
- [9] **Ballian, A.**, Meijers, M. M. J., Methner, K., Cojan, I., Huygue, D., Fiebig, J., and Mulch, A.: A Miocene (23–12.5 Ma) continental paleotemperature record from the northern Mediterranean region (Digne-Valensole Basin, SE France), poster, EGU General Assembly 2023, Vienna, Austria, 23–28 Apr 2023, EGU23-14517, <https://doi.org/10.5194/egusphere-egu23-14517>, 2023.
- [10] **Ballian, A.**, Meijers, M. M. J., Methner, K., Cojan, I., Huygue, D., Fiebig, J., and Mulch, A.: Miocene stable isotope based paleoelevation for the Alps, poster, 6th Annual AlpArray and 4D-MB Scientific Meeting, Hotel Globus, Prague, 12–14 Oct 2022, <http://dx.doi.org/10.17169/refubium-40985>, 2022.
- [11] **Ballian, A.**, Meijers, M. M. J., Methner, K., Cojan, I., Huygue, D., Fiebig, J., and Mulch, A.: A Miocene (23–10 Ma) continental paleotemperature record from the northern Mediterranean region (Digne-Valensole Basin, SE France), poster, 12th International Conference on Climatic and Biotic Events of the Paleogene, University of Bremen, Bremen, 22–25 Aug 2022, 2022.

ADDITIONAL SKILLS

- Languages: French (native), English (C2), German (C1)
- Python programming skills, QGIS, Matlab
- Development of a data visualisation application: <https://clumped.streamlit.app>